

# Born from causality, Tsirelson from steering, and the amplitude seam (SUBMISSION v3)

## Born from causality, Tsirelson from steering, and the amplitude seam:

## an internal reconstruction inside a geometric hidden-connection model

**Cycle 3 draft v1.** Author: A. Oktyabrev (aoktyabrev@gmail.com). Status: assembly draft (~11 pp + battery appendix). All numbers cite raw as file (commit). Claim-discipline per program standard: theorem only with written proof + battery + review; conjecture otherwise.

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### Abstract

Within a geometric hidden-connection (“ribbon”) model we close, and then delimit, an internal reconstruction of quantum correlations. (i) Where a mechanism carries **steering** — conditional states with internally generated diversity  $D(\chi) > 0$  — the no-signaling prohibition forces the Born exponent  $p=2$  uniquely; the derivation is internal (the steering ensembles are produced by a shifted source measure, not imported), executable, and robust under five deformations of the measure. (ii) Mechanisms that resolve outcomes **locally** are capped by Bell’s theorem at  $S=2$ , with a **1/3-cosine point** ( $E=\cos\theta/3$ ) as their isotropic-cosine ceiling, anchored analytically and reached by two mechanisms, with ISO-DYN bracketing it. (iii) The **amplitude seam** between that ceiling and the quantum value  $S=2\sqrt{2}$  is not crossed by decorating local mechanisms — three registered attempts fail on a trilemma of horns (no steering / no amplitude / superluminal telegraph) — but only by **joint, nonlocal resolution**, which by our foliation results demands a shared, statistically invisible precedence structure — a preferred foliation being one realization, a frameless shared order-coin another (class-M theorem). The bridge statement is an exact division of labor, not a unification.

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### 1. Introduction — the import and the question

The program’s canonical text acknowledged an **import**: partially entangled ensembles were introduced by hand (“within this ansatz... we make no claim of a general derivation of the Born rule”, FINAL\_v1:284-290). Cycle 3 asks whether the steering that underlies the Born rule can be **generated internally** by the model’s own

geometry, and — once it is — what exactly stands between such a mechanism and full quantum correlations. The answer assembles into four registered statements (RULE, CEILING, SEAM, FRAME), each with kill-criteria and public commits.

2. Method frame

This work is preregistered. Each campaign registers predictions, estimators, and **kill criteria** (owned by the operator) *before* raw data; the pipeline is raw → commit → analysis, with executable prereg↔JSON assert checklists run before every battery. Batteries are pure numpy (N=2·10<sup>6</sup> unless noted), PRNGKey from config; analytics sit beside numerics (a >2σ mismatch is a stop). Claim-discipline: **theorem** only with a written proof, a battery, and operator review; **conjecture** otherwise.

**Reproduction.** The full numpy battery suite of cycle 3 reproduces bitwise: re-executed with fixed seeds, every results JSON is byte-identical under git diff (C3-B B1-B7, hardening, L3, L1, L2, C3-S, β-coin), and C3-B and the β-coin were independently reproduced by the operator in a separate environment.

**Self-correction register (a feature, not dirty laundry).** Every false stop and retraction below was caught by a preregistered criterion or a literature check, not by hindsight; all are public with commits.

| # | What                                                               | Caught by                                   | Resolution                                                   | Commit  |
|---|--------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------|---------|
| 1 | Δ(2)>0<br>(normalized conditional axis broke total probability)    | “Δ(2)>0 = bug” criterion                    | raw sub-unit Bloch vector                                    | 968aaa5 |
| 2 | Quadrature-tolerance gate (absent from prereg) false-stopped T1/T2 | prereg↔registered criterion                 | <2σ criterion + algebraic zero at p=2; no threshold loosened | ccf5af6 |
| 3 | Monte-Carlo cross-range would false-stop the foliation test        | control design                              | common-random-number comparison                              | 49e8c1b |
| 4 | §2.3 over-claim (“no-signaling alone bounds Tsirelson”)            | battery rediscovered the PR-box fact [PR94] | narrowed to steering-endowed layers                          | 2053106 |

| # | What                                          | Caught by                           | Resolution                                                                       | Commit  |
|---|-----------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------|---------|
| 5 | Frame over-claim (“preferred frame required”) | class-M theorem T3’ + $\beta$ -coin | “shared statistically invisible precedence structure”; foliation one realization | c8e1bf3 |

Section 8 gives the detailed methods and controls; this frame is the summary a reader needs first.

### 3. Model

**Source / steering (F2).** A shared connection orientation  $\lambda$  on  $S^2$  is drawn from a shifted measure  $\mu_\chi \propto 1 + \chi \lambda_z$ . A measurement setting  $a$  partitions  $\lambda$  by  $s = \text{sign}(\lambda \cdot a)$ ; the conditional ensemble’s **raw** (sub-unit) mean vector  $m_s$  is the steered state; readout probability  $f_p(c) = |1 + c|^{\{p/2\}/(|1+c|)} \{p/2\} + |1 - c|^{\{p/2\}} \{p/2\}$ , with  $f_\infty$  the step  $1\{c > 0\}$ . **Lesson (record):** normalizing the conditional axis broke total probability and produced  $\Delta(p=2) > 0$  — the “ $\Delta(2) > 0 = \text{bug}$ ” criterion caught it; the raw sub-unit Bloch vector is correct (battery\_c3b, 968aaa5).

**Joint / chord layer.** A postulated pair law  $P(s, t | a, b) = (1 - st \cos \theta) / 4$  ( $E = -\cos \theta$ ,  $S_{\max} = 2\sqrt{2}$ ), and its weighted generalization  $|s\sqrt{w_a} a - t\sqrt{w_b} b|^p$ , used where the mechanism resolves the pair jointly rather than by local readout of  $\lambda$ .

### 4. RULE — Born from no-signaling, internally (C3-B)

**Steering is generated, not imported.** With  $\mu_\chi$ ,  $\chi > 0$  yields steering diversity  $D(\chi)$ : 0.03→0.08 as  $\chi$  grows (C3B\_hardening.json, 968aaa5) — remote ensembles with **equal means, distinct higher moments**.

**Selection lemma (executable).** For the readout as a random variable over the conditioning, no-signaling  $\Leftrightarrow$  the readout is affine in the projection (Jensen gap of an affine map  $\equiv 0$ ). Affinity  $\cap$  oddness ( $f(-c) = 1 - f(c)$ )  $\cap$  normalization = the single rule  $f_2 = (1 + c)/2 = \mathbf{p=2}$  **uniquely**. The algebraic zero at  $p=2$  is exact (addendum2, ccf5af6); the non-affine tail is strictly positive — scan  $\Delta(4) = 0.042$ ,  $\Delta(6) = 0.105$ ,  $\Delta(10) = 0.217$ ,  $\Delta(\infty) = 0.375$ , monotone, analytic $\leftrightarrow$ numeric  $< 2\sigma$  (C3L\_L3.json, 9567ddb).

**Hardened to theorem.** T1 (sequential/order, B7): repeatability 1.0000, order commute, no-signaling both orders, post-measurement null-set  $\{2\}$ . T2 (anti-circularity): **five** measure deformations ( $\mu_\kappa$ ,  $\kappa \in \{0.5, 2, 3\}$ ; two cubic), all  $D > 0$ , analytic null-set  $\{2\}$ ,  $|\text{analytic} - \text{numeric}| < 2\sigma$ . T3 (F1 direct law) is a boundary (postulated, not steering-collapse). Verdict: **internal derivation of the Born rule from no-signaling within the ribbon’s steering class** (C3B\_hardening.json, c152c98). One **false stop** en route — a quadrature-tolerance gate absent from preregistration — was caught, documented, and corrected without loosening any registered criterion (addendum2, ccf5af6; §8).

**Mechanical realization (C3-B-mech, form-free).** The steering premise is realized mechanically and form-free: an annealed, field-magnetized and fast-frozen

elastic source supplies a polar-biased orientation measure that fits no registered closed form; no closed form is needed. The sign-readout conditionals of that measure carry steering diversity in the third and fifth moments, matching the Jensen analytics computed directly on the empirical ensembles, with the unbiased ( $h=0$ ) control showing null effects throughout; on ensembles produced entirely by the elastic dynamics, no-signaling selects  $p=2$ . The selection theorem's premise thus rests on measured moments, not on an assumed measure — a weaker assumption and hence a stronger claim (C3Bmech\_M1M2.json, 2080bbe;  $M1\checkmark M2\checkmark$  in form-free mode). That the source measure admits no closed form is kin to the C2-TM wall (the relaxational amplitude  $A(k_f)$  likewise has no closed form): an observation — elasticity does not favor closed forms — not a theorem.

## 5. CEILING — the 1/3-cosine point

Local resolution (hidden  $\lambda$  + local linear readout) yields  $E(a,b)=E_{\lambda}[(\lambda \cdot a)(\lambda \cdot b)]=a^T(\frac{1}{3}I)b = \cos\theta/3$  ( $\chi$ -corrections odd  $\Rightarrow$  vanish),  $S=2\sqrt{2}/3 \approx 0.943$  — the isotropic-cosine ceiling of the local layer. The exact convergence is analytic (lever rule, exactly 1/3) and frame-local ( $E(0)=0.333$ ; C3L\_L2.json, 49e8c1b); dynamical isotropization lands NEAR the point:  $\rho_{\cos}=0.311 \pm 0.010$  (cosine-fit, the form-matched estimator; raw C2ISO\_analysis.json: cells/kf4.0/rho\_cos = 0.3113, a043f8f),  $\rho_{\text{model-free}}=0.336$  (same file) — bracketing 1/3 within  $\sim 2\sigma$ . Estimators named per program rule. Anchored analytically and reached by two mechanisms, with ISO-DYN bracketing it  $\Rightarrow$  a structural boundary, the lower bank of the seam.

## 6. SEAM — the trilemma of horns (C3-S)

Can an internal model reach  $D>0$  and  $S=2\sqrt{2}$  together? Registered attempts, ansätze fixed before batteries: - **S-F1** (weighted chord,  $\alpha=\sqrt{1+\chi}$ ,  $\beta=\sqrt{1-\chi}$ ):  $S(\chi)=2\sqrt{2} \cdot \sqrt{1-\chi^2}$  — reaches  $2\sqrt{2}$  only at  $\chi=0$  (symmetry), but is postulated (no continuous measure)  $\Rightarrow D_{\text{joint}} N/A$ . **Horn: no internal steering**. - **S-F2** (shifted source  $\times$  chord/product mixture):  $S(\chi)=\sqrt{2} \cdot (1+\chi/3) \leq 1.886$ , and  $P(s|a,\lambda)=\frac{1}{2} \Rightarrow$  Alice does not update  $\lambda \Rightarrow D_{\text{joint}} \approx 0$ . **Horn: no amplitude / no steering** (C3S.json, 324f097). - **S-F3** (local bias  $\times$  chord kernel,  $|\cdot|$  ansatz with  $\kappa$ ): steers ( $D>0$ ) but the marginal carries Bob-setting dependence at  $O(\kappa\chi)$  — telegraph 0.500 $\rightarrow$ 0.542 (C3S\_F3\_check.py; analytics 2053106). **Horn: superluminal telegraph** — kin to finite-speed no-gos [BPA12].

**Reduction (Bell corollary, not a new theorem).** The internal class =  $\lambda$ -average of local responses = LHV  $\Rightarrow S \leq 2$ . Closing the seam therefore demands **joint, nonlocal resolution**; and joint resolution demands a shared, statistically invisible precedence structure (§7). Attempt count: 3. (C3S\_seam\_reduction.md.)

## 7. FRAME — foliation (C3-L, L1/L2)

On the chord joint law, two resolution mechanics: - **(i) preferred-foliation:**  $S=2\sqrt{2}$  **identical across five foliations** (cross-disc 0.000000 under common random numbers), foliation statistically invisible (C3L\_L2.json, 49e8c1b). - **(ii) frame-local:** space-like  $\rightarrow$  LHV ceiling  $S=0.943$ ; time-like  $A \rightarrow B$ ,  $B \rightarrow A \rightarrow$  reconstruct  $2\sqrt{2}$ . L1: the joint admits an order-invariant realization (existential PASS, factorized witness disc=0.00); the collapse realization's order-dependence (0.0418 with invariant marginals) is evidence for L2c, not pathology (addendum1).

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**Method note:** the foliation-invariance test required common random numbers — a raw cross-range would false-STOP on Monte-Carlo noise (§8).

**Class-M theorem** (status: theorem; canonical statement in `sim/cycle3/C3L_L2c_THEOREM.md`, `c8e1bf3`): in the class M of outcome-definite sequential resolution mechanisms (shared preparation randomness  $\lambda$ , setting-independence  $M4'$ , no retrocausation),  $S > 2$  is achievable **iff** M admits a shared, space-like-transcending, statistically invisible precedence structure. Realizations demonstrated: a preferred foliation (five axes, cross-discrimination 0.000000 under CRN) and a **frameless per-run shared order-coin** ( $S = 2\sqrt{2} \pm 2\sigma$ , no telegraph, coin invisible; battery `c8e1bf3`, independently re-executed). The earlier conjecture's "preferred foliation is required" was **retracted as too strong** during the adversarial passes: the foliation is one realization of the required structure, not its only form. Experimental anchor unchanged: before-before experiments [ZBGT01, SZGS02] exclude the frame-local branch in nature; what survives is joint resolution carrying a shared, hidden precedence structure. Kin: Bohmian preferred slicing [DGZ92], steering-axiomatic reconstructions [CDP11]; [H92], [SS97].

## 8. METHODS — batteries, controls, self-correction (detail; summary in §2)

All batteries: pure numpy,  $N = 2 \cdot 10^6$ , PRNGKey from config, analytics beside numerics ( $> 2\sigma$  mismatch = stop), assert `prereg→JSON` before raw, raw→commit→analysis. Estimator named in every prediction; metric = the model observable.

**Registered false stops and their resolution (part of the evidence):** 1.  **$\Delta(2) > 0$  = bug** (C3-B): normalized conditional axis broke total probability; fixed to raw Bloch (968aaa5). 2. **Quadrature gate** (hardening): an analytic-tolerance gate (tol=1e-9 on a 20001-pt trapezoid, error  $\sim 2.3e-5$ ) false-STOPped T1/T2; corrected to the registered  $< 2\sigma$  criterion + algebraic zero at  $p=2$ ; threshold **not** loosened to pass (addendum2, ccf5af6). 3. **Monte-Carlo cross-range** (L2): raw 5-foliation range would false-STOP; replaced by common-random-number comparison (49e8c1b). 4. **§2.3 over-claim retraction** (C3-S): the first §2.3 asserted "no-signaling alone cuts  $p > 2$  in the joint layer"; J5 rediscovered the Popescu-Rohrlich fact that the postulated  $|\cdot|^p$  family is no-signaling up to the PR box [PR94]; the claim was narrowed to steering-endowed layers (324f097; §9). 5. **Frame over-claim** (abstract/§5/§8/§9 v1): "preferred frame required" — retracted by the class-M theorem's  $T3'$  and the  $\beta$ -coin demonstration; revised in v2, history retained. §9 CEILING wording aligned in v2. Each was caught by a preregistered criterion, not by hindsight.

**Reproduction.** The full numpy battery suite of cycle 3 was independently re-executed (2026-07-21); fixed-seed raw reproduces bitwise (C3-B B1-B7, C3-B hardening, L3, L1, L2, C3-S,  $\beta$ -coin — all results JSON byte-identical under git diff). C3-B and the  $\beta$ -coin were additionally reproduced by the architect in a separate environment.

## 9. The Tsirelson section (§2.3, corrected — verbatim)

Rule selection is exactly as strong as steering. In any layer whose measurements produce conditional states that can themselves be steered — internally generated diversity  $D > 0$  — the Jensen argument applies: no-signaling forces affinity, selects  $p=2$ , and with it the quantum value of  $S$

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available to that layer. In a bare-correlation joint layer, however, where outcomes carry no post-measurement structure, the  $|\cdot\rangle^p$  family has uniform marginals for every  $p$ : the entire family is no-signaling, up to and including the PR box — the classic observation of Popescu and Rohrlich [PR94], which our battery rediscovered as a registered finding against an earlier over-claim of this section (retained in the record). No-signaling alone therefore does not enforce the Tsirelson bound; what enforces it, in this geometry, is no-signaling PLUS steering structure — and this localizes, within one construction, the same gap that information causality [Paw09] was invented to fill. The amplitude seam then resolves into a reduction rather than a mystery: within the internally-generated class (hidden measure plus local response) Bell’s theorem itself caps  $S$  at 2, so closing the seam —  $D>0$  together with  $S=2\sqrt{2}$  — demands joint, non-local resolution of the pair; and joint resolution, by the foliation results above, demands a preferred frame that the statistics provably hide. [Revised by the class-M theorem, §7: read ‘a shared, statistically invisible precedence structure’, of which a preferred frame is one realization — the frameless order-coin is another.] The program’s bridge statement follows: Einstein’s prohibition selects the quantum rule wherever mechanisms carry steering; the quantum amplitude is bought only by joint resolution, whose frame Einstein’s own statistics render invisible.

## 10. Outlook (verbatim)

Three cycles of this program asked one question from three sides: what stands between a mechanism and quantum correlations. The answer assembled itself into four registered statements. (1) RULE: wherever a mechanism carries steering — conditional states with internally generated diversity — the prohibition of superluminal signaling forces the Born exponent  $p=2$  uniquely; the derivation is internal, executable, and robust to deformations of the source measure. (2) CEILING: mechanisms that resolve outcomes locally (hidden measure plus local response) are capped by Bell’s theorem at  $S=2$ , with the  $1/3$ -cosine point as their isotropic-cosine ceiling — a landmark anchored analytically (the lever rule), reached by frame-local resolution, and bracketed by relaxational dynamics (estimator-dependent, §5). (3) SEAM: the gap between that ceiling and the quantum value is not crossed by decorating local mechanisms — our registered attempts fail on a trilemma of horns (no steering, no amplitude, or superluminal telegraph; kin to finite-speed no-gos [BPA12]) — it is crossed only by joint resolution of the pair. (4) FRAME: joint resolution of the frame-local kind reconstructs the Suarez-Scarani alternative and degrades to the local ceiling, which real before-before experiments [ZBGT01, SZGS02] have excluded in nature; the surviving corner, joint resolution carrying a shared, statistically invisible precedence structure (foliation or shared order-coin), reproduces quantum statistics exactly while rendering its own precedence structure statistically invisible to the resolution of our batteries. The bridge statement of the program is therefore not a unification but an exact division of labor: Einstein’s causality selects the quantum rule; the quantum amplitude is bought only by joint resolution; and the precedence structure that joint resolution requires is precisely what the selected statistics forever hide.

What remains open is stated as precisely: a mechanical realization of the asymmetry parameter (C3-B-mech), the structural theorem behind the seam trilemma (the class-M theorem, proved; N-event generalization = C4 G-T), and the ground-state problem of the stiff chain — walls mapped, kill criteria attached, commits public.

**[ARCH placeholder — cycle-4 paragraph, verbatim after  $\gamma$  on T2(N)]** One paragraph: the class-M theorem generalizes to the Mermin *value* at N=3 (theorem, N=3); the tripartite mirror (seam GM-F2 / crossing GM-F2j, the GHZ signature at the Mermin settings); the block-size tier is binary (genuine-N requires a block of size N; Svetlichny  $S_3=4$  at the biseparable bound); the N-partite steering cut is stated as an open problem. Text supplied by the architect after the T2(N) adversarial pass is closed.

**Appendix A — batteries (reproducible)**

| Battery                    | Prereg          | Raw             | Result                                        |
|----------------------------|-----------------|-----------------|-----------------------------------------------|
| C3-B B1-B7                 | dd7a74e         | 968aaa5         | steering generated, p=2 forced                |
| C3-B hardening T1-T3       | 96a3178         | c152c98         | internal derivation (T1✓T2✓)                  |
| L3 scan + bridge           | 2d78c51         | 9567ddb         | tail $\Delta>0$ ; internal $S(2)=\sqrt{2}$    |
| C3-L L1/L2                 | 0b10f5d,100a261 | 1e0cf0a,49e8c1b | foliation invisible; frame-                   |
| C3-S J1-J7                 | 3d516d4         | 324f097         | local→ceiling seam not closed (S-F1/S-F2)     |
| S-F3 analytics             | —               | 2053106         | closed pre-battery (telegraph)                |
| $\beta$ -coin (class M)    | C3L_coin_prereg | c8e1bf3         | frameless $2\sqrt{2}$ , invisible             |
| Class-M theorem            | —               | c8e1bf3         | achievability iff shared invisible precedence |
| C3-B-mech M0 (calibration) | C3Bmech addenda | 0a6c51e         | source polarizes; $\chi(h)$ linear            |
| C3-B-mech M1/M2            | 63456cd         | 2080bbe         | mechanical steering, form-free (M1✓M2✓)       |

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